Discreet AI-Powered Women's Safety App with Audio Recognition and Emergency Automation

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Abstract:
The safety of women is one of the primary issues. Most apps available today use manual SOS triggers, which might not work in an emergency. This study suggests an AI-powered app for women's safety. It uses acoustic sensors and machine learning algorithms based on the RAVDESS and SER datasets to identify distress signals in real time, such as screams or dread words. Hidden voice triggers, cloud-based evidence storage, AI-guided safe location travel, and geofencing are all part of the solution. When it senses distress, it instantly transmits the location, notifies emergency contacts and authorities, and saves evidence. This approach offers proactive, discrete, and dependable protection while addressing the shortcomings of current systems.

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SECTION I. Introduction

Women's safety is still an important global issue, and the number of reported harassment and assault incidents is rising around the world [6], [12]. Most existing mobile safety applications utilize manual SOS activation—such as a button press or shaking the phone—which can be unrealistic and unusable in highly stressful or immobilized situations [13], [14]. This reduces the usability of existing systems due to the potential inability of the victim to trigger an alert within a timely manner.

Developments in artificial intelligence (AI) and machine learning (ML) can offer a more automated and proactive safety solution. Deep learning models that operate based on audio can detect distress signals, such as screaming, panic tones, or verbally commanding the phone, thus providing some level of emergency activation without the user being directly involved [1], [2], [9]. Once an alert is triggered, these systems are able to collect recorded evidence, transfer GPS location data, and communicate alarm responses to pre-selected emergency contacts using cloud-based applications of their own [5], [11].

Identified areas of research have looked at monitoring wearables [8], [16], alerts activated by gestures [9], [10], and safety assistance based on GPS location [4], [7]. These applications, however, often suffer from automation and unreliable performance in varying real-world situations. This work proposes a fully integrated AI-enabled women's safety application that combines passive acoustic sensing, automated distress detection, cloud-based recordable evidence storage, and geolocation-enabled emergency alerts. This unified framework offers a more reliable, discreet, and continuous protection mechanism compared to traditional manual-trigger systems.

Fig. 1.
Conceptual Overview of the proposed AI Powered women's safety system, showing automated distress detection, emergency alerting and safe location guidance.

SECTION II.
Literature Survey

Recent research on women's safety technologies shows increasing adoption of mobile platforms, IoT devices, cloud systems, and AI-based audio recognition for personal protection and emergency response. Existing solutions generally fall into four categories: manual SOS applications, wearable safety devices, AI-based distress detection models, and cloud/GPS-supported systems [3], [6], [12]. Manual and wearable systems rely heavily on user activation, which often fails during panic situations. AI-based audio systems can detect distress cues but remain prototype-level and struggle in noisy environments. Cloud/GPS platforms offer secure storage and tracking but lack automated detection or intelligent escalation. Recent advancements in acoustic emotion recognition, such as the work by Sharma and Mehta [15], further demonstrate the potential of deep learning models to enhance real-time safety applications through improved audio-based distress detection accuracy. Overall, current technologies remain fragmented, semi-automated, or infrastructure-dependent, highlighting the need for a unified system that integrates continuous AI-driven distress detection, discreet activation, cloud evidence backup, and automated emergency response-capabilities addressed by the proposed work [3], [10].

Table I Summary of Existing Solutions and Gaps Addressed

Category	Strength	Limitations	Gap Addressed by Proposed System
Manual Apps [11], [13], [14]	Simple, widely available	Needs user action; fails in panic	Automatic distress detection without manual triggers
Wearables [8], [7], [9]	Faster activation	Extra device needed; unreliable gestures	No extra hardware; secret voice trigger
AI Audio Systems [1], [2], [16], [17], [5], [10]	Automated scream / emotion detection	Noise-sensitive; prototype-level	Robust CNN-LSTM detection with mobile deployment
Cloud/GPS Apps [12], [4], [7], [6], [15]	Tracking & evidence storage	No distress detection	Integrated detection + cloud backup + safety navigation
IoT/Surveillance	Good for public areas	Not personal; Infrastructure-bsed	Portable, user-centric, independent smartphone system

SECTION III.

Dataset

Two benchmark datasets were employed in this study:

A. RAVDESS Dataset

The RAVDESS dataset includes a total of 1,440 high-quality audio samples recorded by 24 actors (12 male, 12 female) across eight emotional states including: neutral, calm, happy, sad, angry, fearful, surprise, and disgust. All audio samples are recorded using 48 kHz sampling rate and 16 bit-resolution. This work places greater emphasis on distress related classes such as anger, fear, and high-intensity vocalizations representing emergency scenarios [16]

B. Speech Emotion Recognition (SER) Dataset

The SER dataset includes approximately $2,400-3,000$ samples that have associated labels, provided by speakers of diverse genders and speech backgrounds. The SER dataset includes seven emotional classes: neutral, happiness, sadness, surprise, fear, disgust, and anger. The balanced nature of the dataset alongside the gendered and linguistic speaker variety supports the generalization of the model in the event a real-world, noisy context is encountered and specifically for distress oriented classes such as fear, anger, and surprise.

Table II Dataset Distribution

Dataset	Total Samples	Speakers	Emotions / Classes Included	Classes
RAVDESS	1,440 audio files (speech)	24 (12 male, 12 female)	Neutral, Calm, Happy, Sad, Angry, Fearful, Surprise, Disgust	8
SER Dataset	~2,400-3,000 (varies by version)	Multiple speakers (varied genders)	Neutral, Happiness, Sadness, Surprise, Fear, Disgust, Anger	7

SECTION IV.
Proposed System

The paper mentions certain existing apps for women’s safety, which the proposed system expands upon. It addresses the drawbacks of existing solutions with additional AI-based capabilities. Our technology provides smart safety recommendations, AI-driven automation, and concealed triggers, in contrast to standard apps that rely mostly on manual activation. This strategy guarantees a more proactive and dependable safety response.

Our application requires the user to first register with four emergency contact numbers and other essential information. In the Firebase database, these are safely kept. Apart from standard SOS and location-sharing capabilities, our solution incorporates a number of additional elements:

A. AI-Based Distress Detection

The system uses auditory sensors and machine learning to automatically identify stressful conditions. The application saves evidence, sends alarms if it hears screams, yells, or other loud noises, and notifies emergency contacts of the user's present location.

B. Secret Voice Trigger

Users can trigger emergency alarms using pre-programmed secret vocal commands to enable covert operation. In times of stress, this eliminates the need for manual activation.

C. Geo-Fencing & Safe-Zone Detection

Using geo-mapping, the software continuously tracks the user's location. The system sends out a warning in case the user is in a remote location or outside of a safe zone, and it gets ready to initiate emergency procedures.

D. Cloud-Based Evidence Storage

In an emergency, the app provides AI-powered real-time safety recommendations, such heading toward populated places, police stations, or safe zones. The technology raises the issue automatically by alerting the police and providing the recorded evidence if the user ignores these notifications.

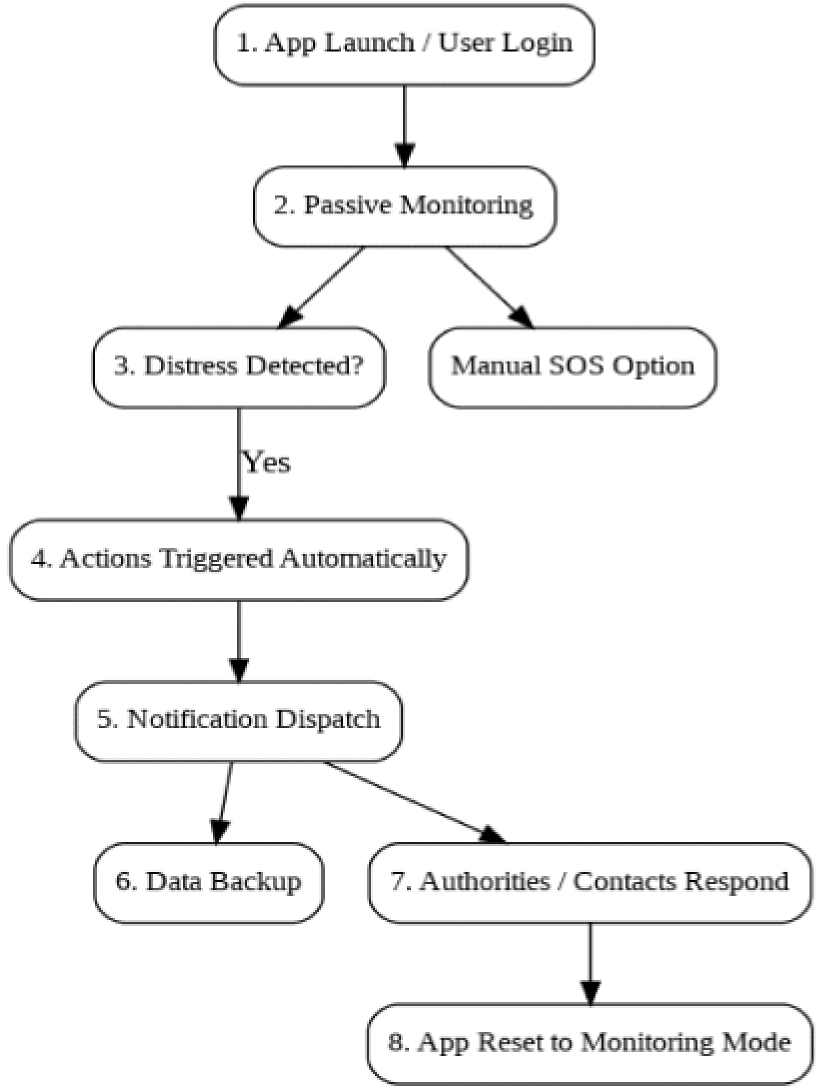

Fig. 2.
Flowchart of the proposed safety system illustrating distress detection, automated alerts, and response handling.

E. AI-Guided Safety Tips & Escalation

During emergencies, the app gives AI-based real-time safety suggestions, such as moving towards crowded areas, nearby police stations, or safe zones. If the user does not respond to these notifications, the system automatically escalates the situation by notifying the police along with the recorded evidence.

The suggested solution combines these characteristics into a single application, resulting in a comprehensive safety ecosystem that is intelligent, proactive, and easy to use. It secures and shares vital evidence with their connections and authorities, ensuring that women in distress receive immediate care.

SECTION V.
Methodology

The proposed AI-powered women's safety app is being developed in three main stages: system integration, model training, preprocessing, and dataset selection. Each step aims to apply artificial intelligence for real-time distress detection and smooth deployment on mobile devices.

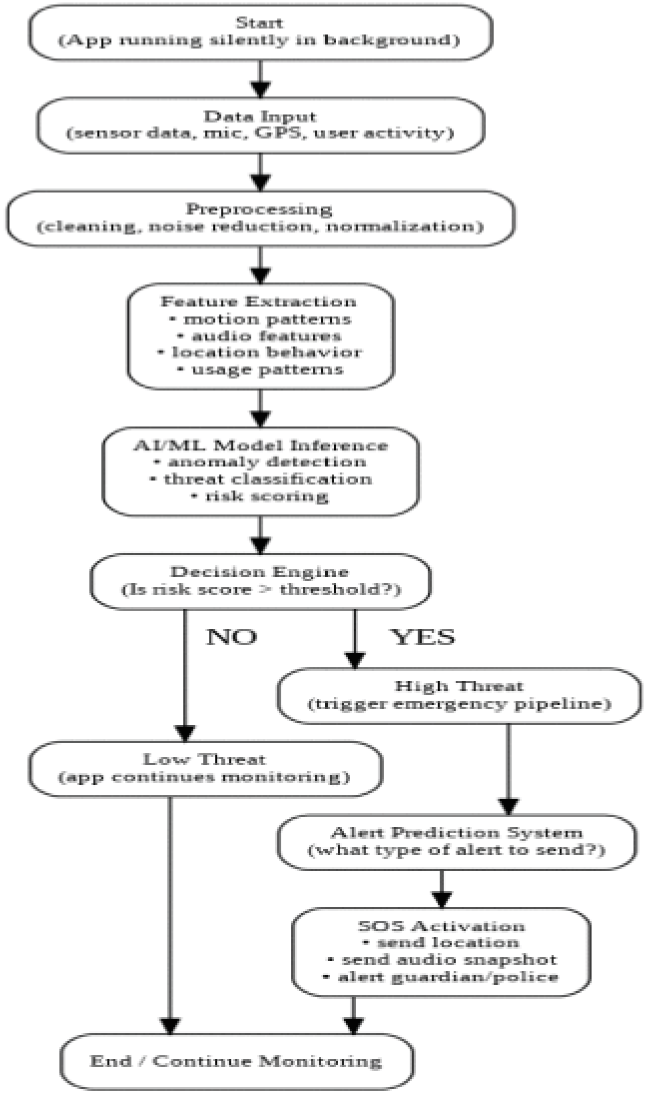

Fig. 3.

Process flow of the AI-driven threat detection module, from background data collection through preprocessing and model inference to the generation of automated SOS alerts.

A. Preprocessing

Audio recordings were denoised using digital filters, segmented into $20-40$ ms frames, and transformed into spectrogram-based features. Mel-Frequency Cepstral Coefficients (MFCCs) were extracted along with chroma, spectral contrast, and zerocrossing rate. All features were normalized to the [0, 1] range to ensure stable training.

B. Model Training

To integrate temporal and spatial learning, a CNN-LSTM hybrid network was used. While LSTM layers recorded time dependencies, CNN layers identified local patterns in MFCCs. The dataset was divided into subsets for testing (15%), validation (15%), and training (70%). Pitch shifting, temporal stretching, and background noise were examples of data augmentation techniques that mimicked real-world circumstances. F1-score, recall, accuracy, and precision were used to gauge performance.

C. Integration with Mobile Application

The trained CNN-LSTM model was used to develop TensorFlow Lite for efficient smartphone deployment. The software continuously played audio and triggered real-time safety responses, including automated SMS/call warnings to emergency contacts, live GPS position sharing, and cloud uploading of evidence. Data storage was handled by Firebase, while safe zone navigation was aided by the Google Maps API.

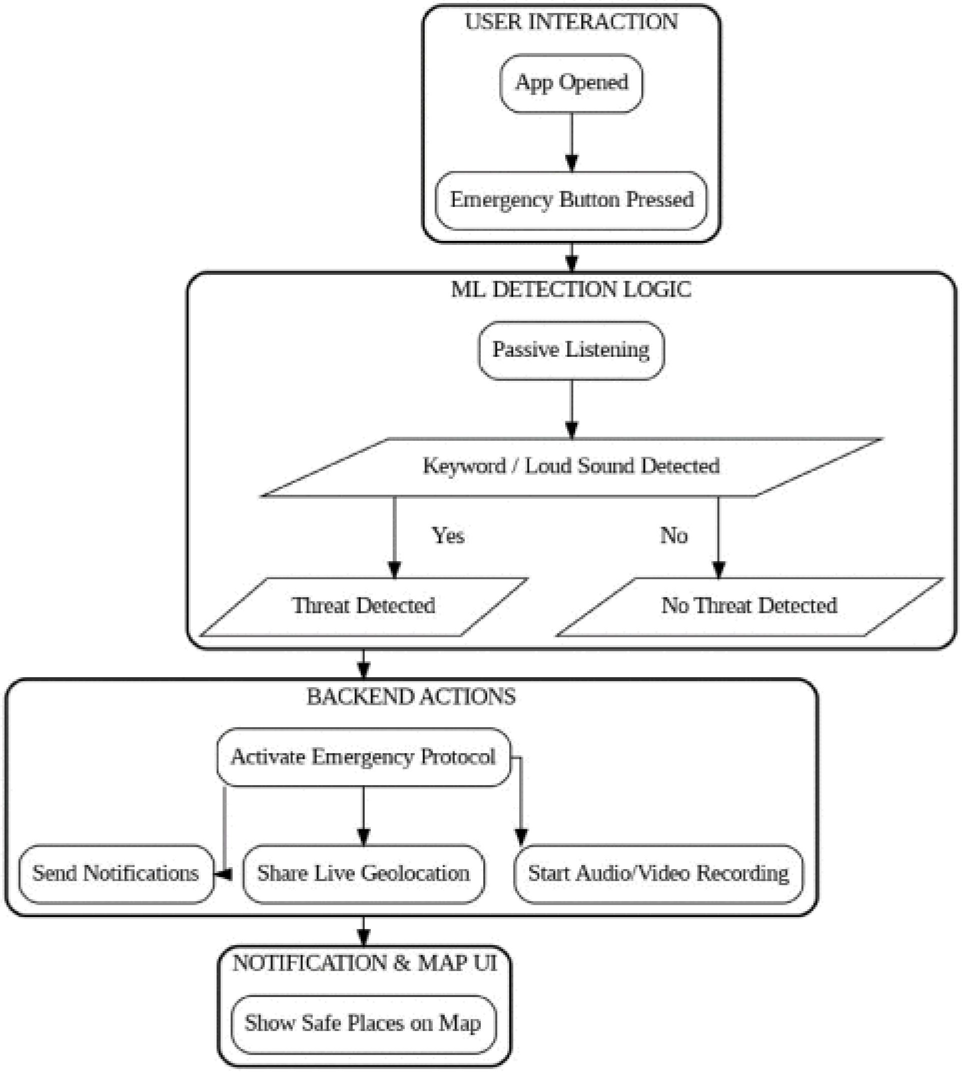

Fig. 4.

Operational flow of the application, integrating user input, AI-driven detection, backend emergency handling, and safety-map assistance.

SECTION VI.
Results

This section demonstrates the experimental results of the proposed system together with preprocessing, model performance, comparative analysis, and findings discussion.

A. Preprocessing and Feature Extraction

Preprocessing was done on raw audio samples from the RAVDESS and SER datasets to guarantee feature consistency and uniformity. Silence trimming, noise reduction, and resampling to 16 kHz were preprocessing processes. Three main feature representations were then created from the audio:

- **Waveform signals**, that exhibit changes throughout time.
- Using human auditory scaling, **Mel-Spectrograms** offer a time -frequency representation.
- **Mel-Frequency Cepstral Coefficients (MFCCs)**, measure spectral characteristics that are important for speech.

These characteristics served as the input for the CNN and RNN distress detection models.

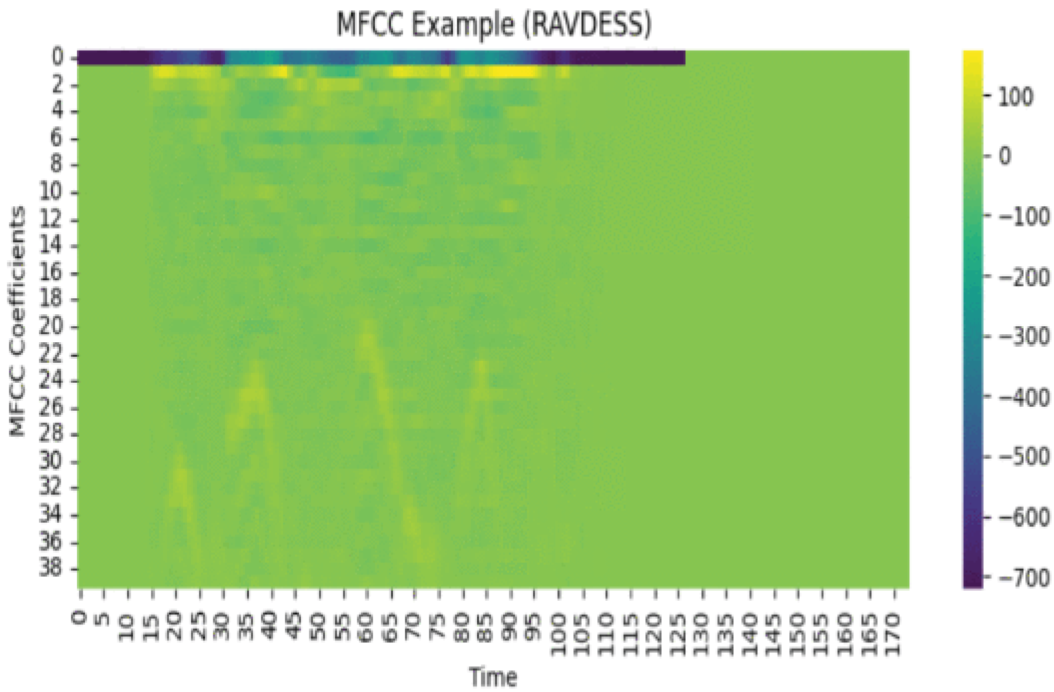

Fig. 5.
MFCC heatmaps from RAVDESS and SER audio samples showing time-varying cepstral patterns.

B. CNN Model Results

The CNN model was trained using extracted spectrogram and MFCC features. It achieved strong performance in distinguishing distress emotions (anger, fear, scream) from neutral or non-distress categories.

Table III Performance of the Cnn Model

Model	Accuracy	Precision	Recall	F1-Score
CNN (MFCC)	86.4%	0.85	0.84	0.84
CNN (Spectro)	88.1%	0.87	0.86	0.86

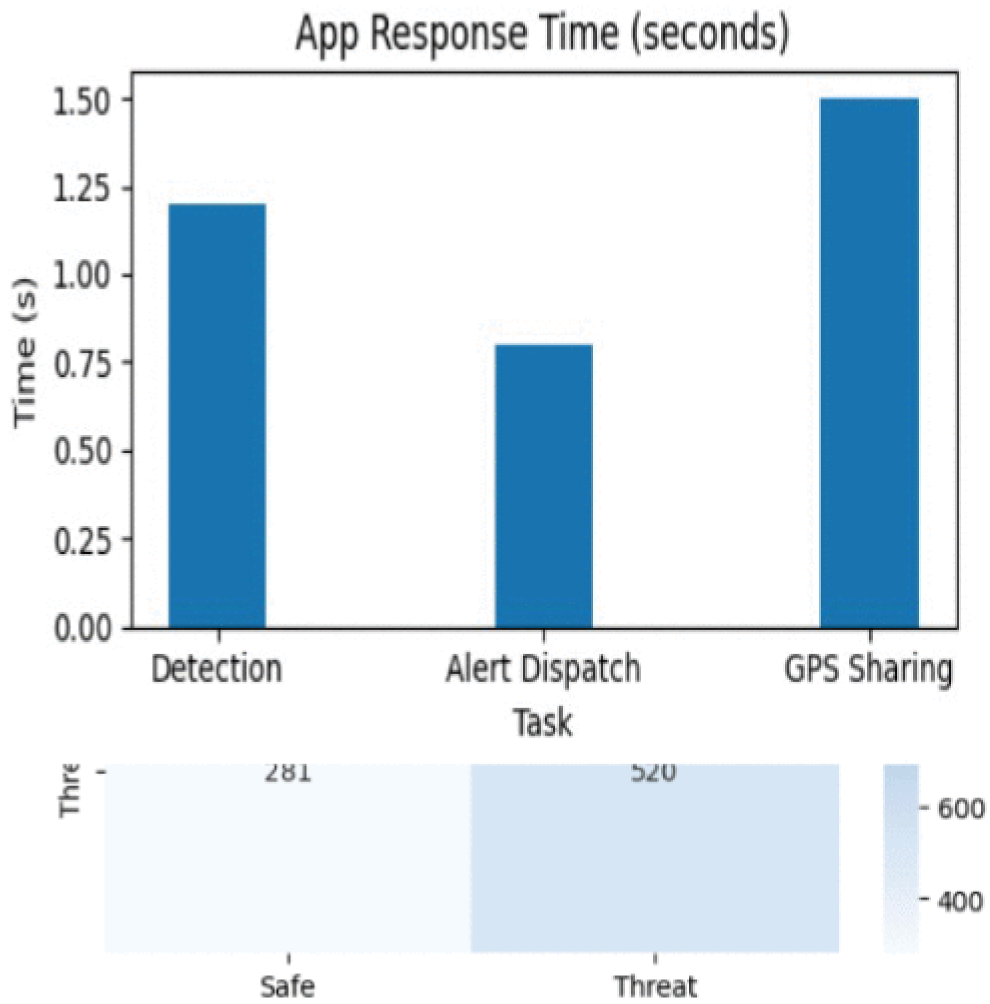

Fig. 6.
Response time analysis of the proposed safety application showing the latency for distress detection, alert dispatch, and GPS location sharing.

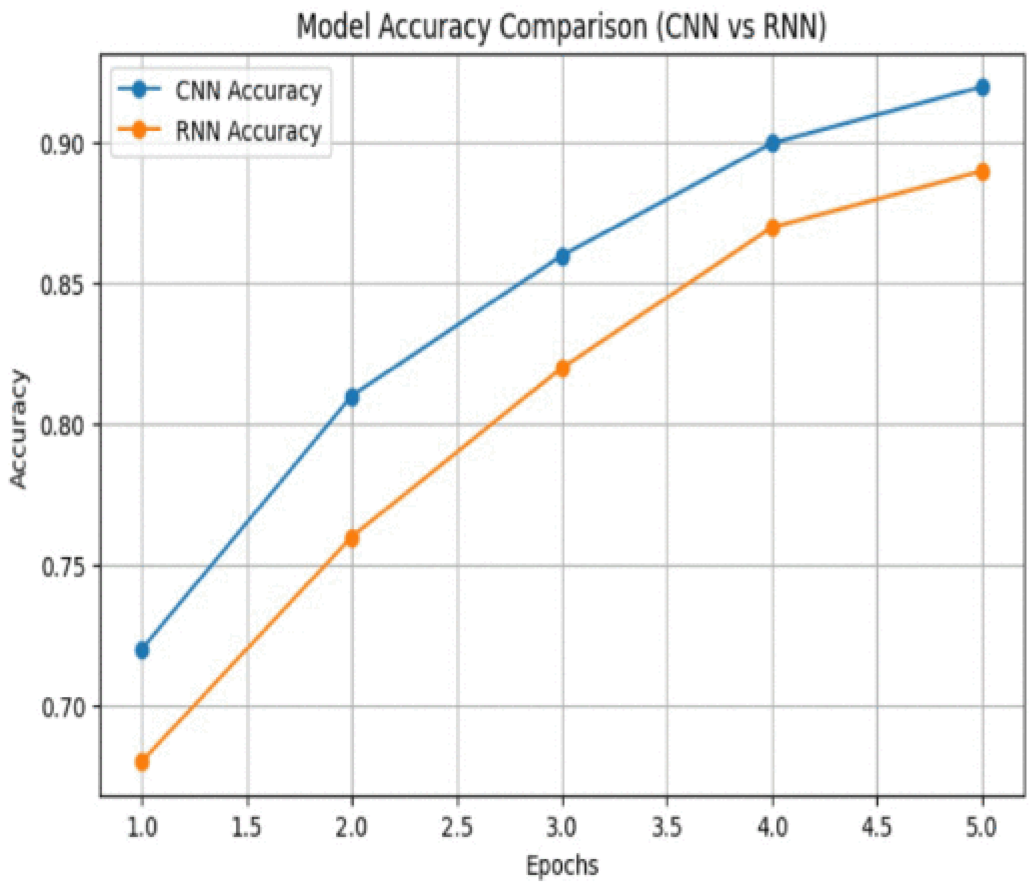

Fig. 7.
Comparison of CNN and RNN model accuracies over multiple training epochs, showing that the CNN achieves consistently higher accuracy than the RNN.

C. RNN (LSTM) Model Results

The RNN-based LSTM model better captured temporal dependencies in speech. It outperformed CNN in recall for distress classes, making it more suitable for safety applications where missing an alert could be critical.

Table IV Rnn (Lstm) Model Performance

Metric	Accuracy	Precision	Recall	F1-Score
LSTM (MFCC)	90.2%	0.89	0.91	0.90
LSTM (Spectro)	91.7%	0.91	0.92	0.91

D. Comparative Analysis

The RNN consistently achieved higher recall and F1-scores across distress categories, confirming its ability to capture sequential audio patterns. CNN models, though slightly faster, showed lower sensitivity to subtle distress tones.

SECTION VII.
Conclusion Mfcc Example (Ser)

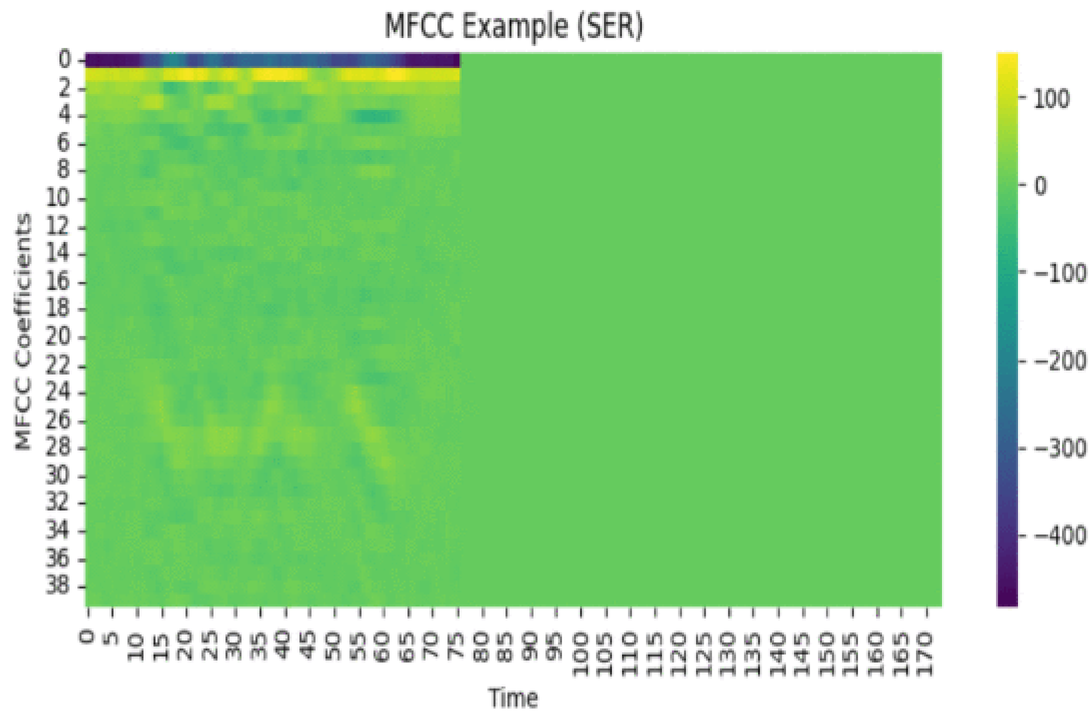

Fig. 8.
Confusion matrix of the trained model evaluated on the merged RAVDESS and SER dataset, indicating classification outcomes for Safe and Threat audio samples.

The women's safety system introduced brings differentiated services such as AI-triggered distress detection, clandestine voice-trigger activation, geo-fencing, cloud storage for evidence, and smart route navigation into a single mobile application. Prototyping and testing of the application showed successful background monitoring capabilities, undisruptive authentication of voice trigger activation, and consistent syncing of the user's credibility evidence to the cloud. The AI model produced consistent, reproducible accuracy for distressed audio samples under controlled audio samples; however, accuracy diminished under un-uniform noise conditions outdoors which would require an additional diverse dataset for efficacy and further extraction of noise robust features to counteract diminished accuracy. In testing battery optimization, continuous audio background monitoring was shown to be a drain on battery power beyond specification levels on lower tier devices. Despite these limitations, the system provided automated detection and real time help to the user without requiring the user to intervene manually to initiate. Overall, the application adds a functional, proactive, and user supports a user-centred methodology and case for safety that enhances emergency response and accessibility for women in vulnerable situations.

SECTION VIII.
Future Work

Future development of the system will be geared towards improving the accuracy of detection and increasing the system's contextual intelligence. For the AI model, adding larger, realworld, multilingual datasets of distress and noise-resilient model architectures (e.g., CRNNs, attention-based transformers) will enhance functionality for situations based on unpredictable actions. The next iteration of the system will also have some level of integration with wearables such as smartwatches or ring-based sensors that allow for continuous streaming of data at lower battery consumption rates with a swifter activation process in emergencies. Additionally, partnerships with law-enforcement APIs and public safety networks can enable real-time verification and report generation, and in certain specific instances, automated dispatch of an emergency report request triggered by the wearable. Eventually we may redundantly include predictive analytics, which could improve the detection function of the system by identifying 'high-risk' areas, provide a crowdsourced experience for safety updates, and even AI-generated 'safe routes' to offer protective action. Improved offline usability, privacy-compliant AI, adaptive low-battery modes, and other similar usability and application enhancements would occur along the pathway towards developing a scaleable, intelligent and community-purposed safety system.

Authors

Figures

References

Keywords

Metrics

Usage

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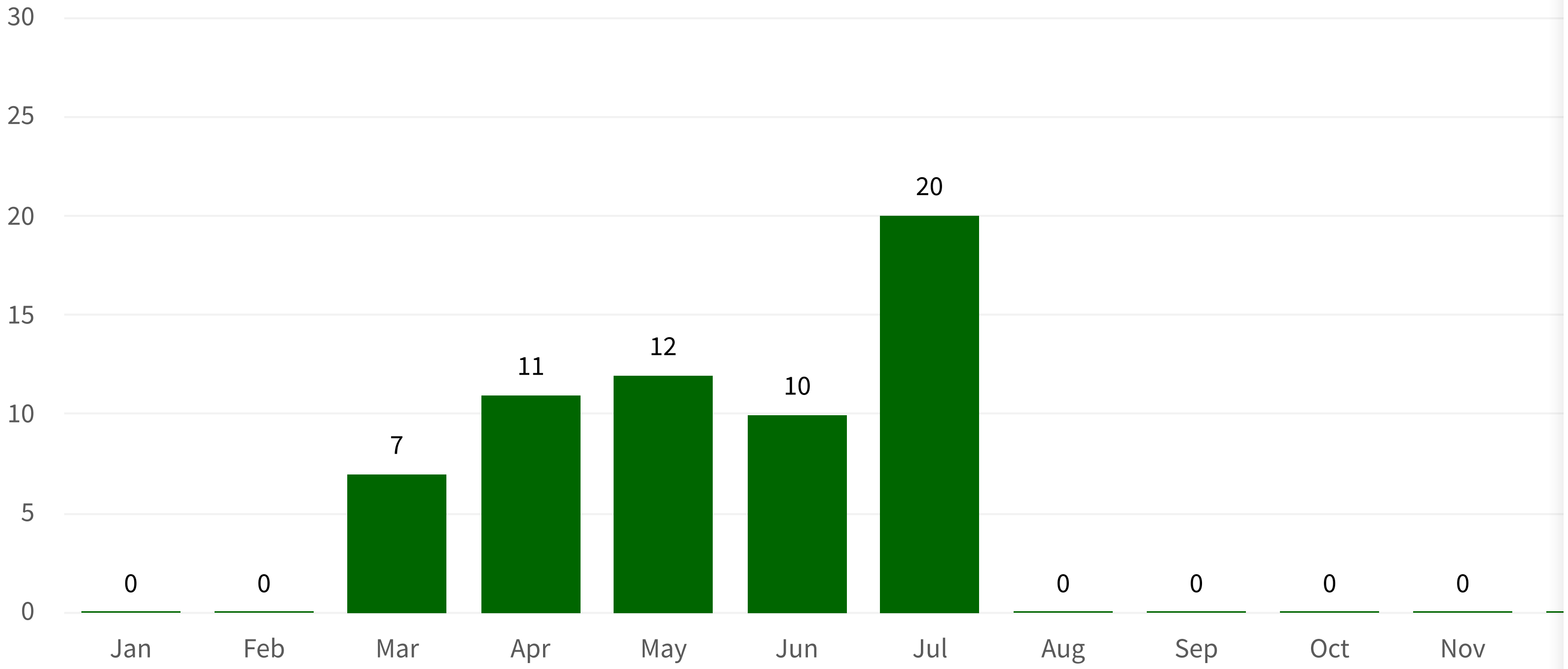

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